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MINI PROJECT REPORT

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INTRODUCTION

Machine Learning is a rapidly growing branch of Artificial Intelligence that empowers systems to learn from historical data and make accurate predictions or decisions without being explicitly programmed for each scenario. In recent years, machine learning has found widespread application in the field of agriculture, where it is being used to address one of humanity's most critical challenges — ensuring food security and improving agricultural productivity.

Agriculture is the backbone of developing economies, and crop yield prediction plays a central role in planning, resource allocation, and decision-making for farmers, government agencies, and agribusinesses. Traditional methods of yield prediction rely on expert knowledge and historical averages, which are often inaccurate and fail to account for the complex interactions between environmental, soil, and agronomic factors. Machine learning offers a powerful alternative by identifying hidden patterns in large datasets and producing reliable, data-driven predictions.

In this project, a regression-based machine learning system is developed to predict crop yield in tons per hectare based on a set of agricultural features including region, soil type, crop variety, rainfall, temperature, fertilizer usage, irrigation type, weather condition, and days to harvest. The system implements and compares four regression algorithms — Linear Regression, Decision Tree Regressor, Random Forest Regressor, and K-Nearest Neighbors Regressor — to evaluate their performance and select the best model. Additionally, K-Means Clustering is applied to discover natural groupings within the agricultural data, providing further insight into productivity zones and crop growing conditions.

PROBLEM STATEMENT

To develop a machine learning system that accurately predicts crop yield in tons per hectare using agricultural and environmental features, by training and comparing multiple regression models, and to identify patterns in agricultural conditions through unsupervised clustering, with the goal of supporting better crop planning and farm management decisions.

GOAL

The primary goal of this project is to build an accurate and reliable regression model that can predict the expected crop yield given a set of agricultural input features. By training multiple machine learning models on labeled agricultural data, the system aims to evaluate each model's predictive capability and identify the best-performing approach based on standard evaluation metrics such as R^2 Score, Mean Squared Error (MSE), and Mean Absolute Error (MAE).

A further objective of this project is to enable model comparison across four distinct regression algorithms, namely Linear Regression, Decision Tree Regressor, Random Forest Regressor, and KNN Regressor. This comparison provides a deeper understanding of how different algorithmic strategies handle the complexity of agricultural data and helps justify the selection of the final model.

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The project also aims to apply K-Means Clustering as an unsupervised learning technique to identify natural groupings in agricultural conditions. These clusters help classify regions or farm settings into productivity zones such as High Productivity, Low Productivity, and Moderate Conditions, providing actionable insights beyond what regression alone can offer.

Finally, the project seeks to demonstrate the practical value of machine learning in precision agriculture by producing visualizations including model comparison bar charts, actual versus predicted graphs, feature importance plots, and clustering scatter plots that communicate results clearly and effectively.

THEORETICAL BACKGROUND

Overview of the Problem

Crop yield prediction is a regression problem in machine learning, where the goal is to predict a continuous numerical output — crop yield in tons per hectare — based on a set of input features. Unlike classification tasks that assign discrete labels, regression models learn the functional mapping between input variables and a continuous target. The challenge in agricultural yield prediction lies in the fact that crop output depends on a complex, non-linear combination of soil properties, weather patterns, irrigation practices, fertilizer usage, and crop type, making it necessary to evaluate multiple model types to find the most suitable one.

Linear Regression

Linear Regression is the simplest and most interpretable regression model. It assumes a linear relationship between the input features and the target variable, and models that relationship using the equation:

$$\hat{y} = w_0 + w_1x_1 + w_2x_2 + \dots + w_nx_n$$

Where $\hat{y}$ is the predicted yield, x_1 through x_n are the input features, and w_0 through w_n are the learned coefficients. The model is trained by minimizing the Mean Squared Error between predicted and actual values. While Linear Regression serves as a useful baseline, it may underperform when the underlying relationships in the data are non-linear, which is common in agricultural datasets.

Decision Tree Regressor

A Decision Tree Regressor builds a tree-like model of decisions by recursively splitting the dataset based on feature values that minimize the variance of the target variable within each split. Each internal node represents a decision based on a feature threshold, and each leaf node represents a predicted yield value. Decision Trees are capable of capturing non-linear patterns and are easy to interpret visually. However, deep trees tend to overfit the training data, which is why the maximum depth is limited during training.

Random Forest Regressor

Random Forest is an ensemble learning method that constructs a large number of decision trees during training and averages their predictions to produce the final output. This approach reduces overfitting significantly compared to a single decision tree and improves generalization on unseen data. The prediction of a Random Forest with T trees is given by:

$$\hat{y} = (1/T) \sum h_t(x)$$

Where $h_t(x)$ is the prediction of the t -th tree. Random Forest also provides feature importance scores, which indicate which input variables contribute most to the prediction. This makes it both a powerful predictive model and a useful tool for understanding the data.

K-Nearest Neighbors Regressor

KNN Regressor is an instance-based learning algorithm that makes predictions by finding the K training samples closest to a new input point and averaging their target values. It does not build an explicit model but instead memorizes the training data and makes predictions at query time. The distance between data points is typically measured using Euclidean distance:

$$d(x, x_i) = \sqrt{\sum (x_j - x_{ij})^2}$$

KNN is effective when local patterns in the data are meaningful, but it is sensitive to the scale of features and the choice of K , and can be slow on large datasets.

K-Means Clustering

K-Means is an unsupervised learning algorithm that partitions the data into K clusters by minimizing the within-cluster sum of squared distances from each point to its cluster centroid. The algorithm iteratively assigns each data point to the nearest centroid and updates the centroids until convergence. The objective function minimized is:

$$J = \sum_k \sum_{x \in C_k} \|x - \mu_k\|^2$$

Where μ_k is the centroid of cluster k . In this project, K-Means is applied to group agricultural records into three clusters based on rainfall, temperature, days to harvest, and yield, revealing natural productivity zones in the data.

Evaluation Metrics

Since this is a regression problem, the models are evaluated using the following metrics. R^2 Score, also called the coefficient of determination, measures what fraction of the variance in the target variable is explained by the model. A value of 1.0 indicates a perfect fit, while a value of 0 means the model is no better than predicting the mean. Mean Squared Error (MSE) measures the average squared difference between actual and predicted values, penalizing large errors more heavily. Mean Absolute Error (MAE) measures the average absolute difference and is easier to interpret in the original units of the target variable.

Literature Survey

Several studies have demonstrated the effectiveness of machine learning in agricultural yield prediction. Linear regression and its variants have been used as baseline models due to their simplicity and interpretability. However, they are often unable to capture the non-linear dynamics of crop growth influenced by environmental factors. Decision tree-based models have been found to handle non-linearity more effectively, and ensemble methods such as Random Forests have consistently demonstrated superior predictive accuracy across agricultural datasets in multiple research works.

Studies by researchers in the domain of precision agriculture have shown that ensemble methods reduce prediction error significantly when compared to individual learners. The use of K-Means Clustering for agricultural data segmentation has also been explored, with researchers identifying distinct farming zones based on climatic and soil characteristics. These findings support the inclusion of multiple models and clustering in the design of this project, ensuring both predictive accuracy and exploratory insight.

Justification for Choosing the Algorithms

Linear Regression is included as the baseline model because it is simple, fast, and provides a reference point against which the performance of more complex models can be measured. Decision Tree Regressor is chosen for its ability to capture non-linear patterns in the agricultural data and for its visual interpretability. Random Forest Regressor is selected as the primary model because it combines the strengths of multiple decision trees, reduces overfitting through ensemble averaging, and provides feature importance scores that add interpretability to the model. KNN Regressor is included to represent an instance-based learning approach and to evaluate how locality-sensitive predictions compare to model-based approaches. K-Means Clustering is applied as a complement to the regression analysis, enabling unsupervised discovery of natural groupings in the data that provide additional agronomic insights.

ALGORITHM EXPLANATION WITH EXAMPLE

Step 1 — Input Data

The dataset used in this project contains agricultural records with the following features: Region (categorical), Soil Type (categorical), Crop (categorical), Rainfall in mm (numeric), Temperature in Celsius (numeric), Fertilizer Used (Boolean), Irrigation Used (Boolean), Weather Condition (categorical), Days to Harvest (numeric), and Yield in tons per hectare (numeric target variable).

Step 2 — Preprocessing

Before the data can be used for training, it must be prepared. Categorical columns such as Region, Soil Type, Crop, and Weather Condition are encoded into numeric form using Label Encoding, which assigns a unique integer to each category. Boolean columns Fertilizer Used and Irrigation Used, which appear as true/false string values, are mapped to 1 and 0 respectively. All

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features are then standardized using StandardScaler to ensure that no single feature dominates the model due to differences in scale. The processed data is split into 80% training and 20% testing subsets using a fixed random seed for reproducibility.

Step 3 — Feature and Target Split

The target variable Yield_tons_per_hectare is separated from the input features, and all remaining columns are used as predictors. This gives the model a complete picture of the farming conditions to learn from.

Step 4 — Model Training

Each of the four regression models is initialized with appropriate hyperparameters. The Decision Tree is limited to a maximum depth of 6 to prevent overfitting. The Random Forest is built using 100 estimators. The KNN model uses 5 nearest neighbors. All models are trained on the same training set to ensure a fair comparison.

Step 5 — Prediction and Evaluation

After training, each model generates predictions on the test set. The R^2 Score, MSE, and MAE are calculated for each model and compared in a summary table. This allows direct identification of the best-performing model.

Step 6 — Clustering

K-Means Clustering is applied on four features — Rainfall, Temperature, Days to Harvest, and Yield — using three clusters. The clusters are automatically labeled based on their average yield values as High Productivity Zone, Low Productivity Zone, and Moderate Conditions. This provides a meaningful agronomic interpretation of the natural groupings in the data.

Mathematical Representation

Random Forest prediction: $\hat{y} = (1/100) \sum_t h_t(x)$

KNN prediction: $\hat{y} = (1/K) \sum_{i \in N_k} y_i$

K-Means objective: $J = \sum_k \sum_{x \in C_k} \|x - \mu_k\|^2$

R^2 Score: $R^2 = 1 - (SS_{\text{res}} / SS_{\text{tot}})$

MSE: $MSE = (1/n) \sum (y_i - \hat{y}_i)^2$

IMPLEMENTATION CODE AND DATASET USED

Category	Description
Dataset Name	Crop Yield Dataset
Domain	Agriculture/Precision Farming
Objective	Regression (Yield Prediction)
Data Source	kaggle
Features	Region, Soil Type, Crop, Rainfall, Temperature, Fertilizer Used, Irrigation Used, Weather Condition, Days to Harvest
Feature Type	Numeric, Categorical, Boolean
Target Variable	Yield_tons_per_hectare
Task Type	Supervised Regression + Unsupervised Clustering

Python code:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings("ignore")

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.linear_model import LinearRegression
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor
from sklearn.neighbors import KNeighborsRegressor
from sklearn.cluster import KMeans
from sklearn.metrics import r2_score, mean_squared_error, mean_absolute_error

df = pd.read_csv('/content/drive/MyDrive/dataset/crop_yield.csv')
print("Shape:", df.shape)
print(df.head())
print("Missing values:", df.isnull().sum())

df_enc = df.copy()

df_enc["Fertilizer_Used"] = df_enc["Fertilizer_Used"].map(
    {True: 1, False: 0, "true": 1, "false": 0})
df_enc["Irrigation_Used"] = df_enc["Irrigation_Used"].map(
```

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```
{True: 1, False: 0, "true": 1, "false": 0})

le = LabelEncoder()
for col in ["Region", "Soil_Type", "Crop", "Weather_Condition"]:
    df_enc[col] = le.fit_transform(df_enc[col].astype(str))

if "index" in df_enc.columns:
    df_enc.drop("index", axis=1, inplace=True)

X = df_enc.drop("Yield_tons_per_hectare", axis=1)
y = df_enc["Yield_tons_per_hectare"]

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42)

print(f"Train: {len(X_train)} | Test: {len(X_test)}")
models = {
    "Linear Regression": LinearRegression(),
    "Decision Tree": DecisionTreeRegressor(max_depth=6, random_state=42),
    "Random Forest": RandomForestRegressor(n_estimators=100, random_state=42),
    "KNN Regressor": KNeighborsRegressor(n_neighbors=5),
}

results = {}
predictions = {}

for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    predictions[name] = y_pred
    results[name] = {
        "R2": r2_score(y_test, y_pred),
        "MSE": mean_squared_error(y_test, y_pred),
        "MAE": mean_absolute_error(y_test, y_pred),
    }
    print(f"{name} → R²: {results[name]['R2']:.4f} | "
          f"MSE: {results[name]['MSE']:.4f} | MAE: {results[name]['MAE']:.4f}")

results_df = pd.DataFrame(results).T.sort_values("R2", ascending=False)
print("\n—— Model Comparison ——")
print(results_df.round(4))

cluster_features = df[["Rainfall_mm", "Temperature_Celsius",
```


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```
"Days_to_Harvest", "Yield_tons_per_hectare"]]  
cs = StandardScaler()  
cluster_scaled = cs.fit_transform(cluster_features)  
  
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)  
df["Cluster"] = kmeans.fit_predict(cluster_scaled)  
  
summary = df.groupby("Cluster") [  
    ["Rainfall_mm", "Temperature_Celsius",  
     "Days_to_Harvest", "Yield_tons_per_hectare"]].mean().round(2)  
print("\n— Cluster Summary —")  
print(summary)  
  
low = summary["Yield_tons_per_hectare"].idxmin()  
high = summary["Yield_tons_per_hectare"].idxmax()  
mid = [c for c in range(3) if c not in [low, high]][0]  
  
label_map = {low: "Low Productivity Zone",  
             high: "High Productivity Zone",  
             mid: "Moderate Conditions"}  
df["Cluster_Label"] = df["Cluster"].map(label_map)  
print(df["Cluster_Label"].value_counts())  
  
plt.figure(figsize=(8, 5))  
colors = ["#2ecc71" if r > 0.85 else "#f39c12" if r > 0.7 else "#e74c3c"]  
    for r in results_df["R2"]]  
plt.barh(results_df.index, results_df["R2"], color=colors, edgecolor="black")  
plt.xlabel("R2 Score")  
plt.title("Model Comparison — R2 Score")  
plt.xlim(0, 1.05)  
for i, v in enumerate(results_df["R2"]):  
    plt.text(v + 0.01, i, f"{v:.4f}", va="center", fontweight="bold")  
plt.tight_layout()  
plt.savefig("model_comparison.png", dpi=150)  
plt.show()  
  
y_pred_rf = predictions["Random Forest"]  
plt.figure(figsize=(7, 6))  
plt.scatter(y_test, y_pred_rf, alpha=0.5, color="#3498db",  
            edgecolors="black", linewidths=0.3)  
plt.plot([y_test.min(), y_test.max()],  
         [y_test.min(), y_test.max()], "r--", lw=2, label="Perfect Fit")  
plt.xlabel("Actual Yield (tons/ha)")  
plt.ylabel("Predicted Yield (tons/ha)")  
plt.title("Actual vs Predicted — Random Forest")  
plt.legend()
```

```
plt.tight_layout()
plt.savefig("actual_vs_predicted.png", dpi=150)
plt.show()

rf_model = models["Random Forest"]
importances = pd.Series(
    rf_model.feature_importances_, index=X.columns).sort_values(ascending=True)
plt.figure(figsize=(8, 6))
importances.plot(kind="barh", color="#9b59b6", edgecolor="black")
plt.title("Feature Importance — Random Forest")
plt.xlabel("Importance Score")
plt.tight_layout()
plt.savefig("feature_importance.png", dpi=150)
plt.show()

color_map = {"Low Productivity Zone": "#e74c3c",
             "High Productivity Zone": "#2ecc71",
             "Moderate Conditions": "#3498db"}
plt.figure(figsize=(8, 6))
for label, grp in df.groupby("Cluster_Label"):
    plt.scatter(grp["Rainfall_mm"], grp["Yield_tons_per_hectare"],
                label=label, alpha=0.6, color=color_map[label],
                edgecolors="black", linewidths=0.2)
plt.xlabel("Rainfall (mm)")
plt.ylabel("Yield (tons/ha)")
plt.title("K-Means Clustering — Agricultural Zones")
plt.legend()
plt.tight_layout()
plt.savefig("clustering.png", dpi=150)
plt.show()
```

OUTPUT

FIGURE 1 — Model Comparison Bar Chart

The horizontal bar chart displays the R^2 Score of all four regression models side by side. Random Forest Regressor achieves the highest R^2 Score, confirming its position as the best model among the four. Decision Tree follows, while Linear Regression and KNN Regressor show comparatively lower scores. Green bars indicate high-performing models, orange indicates moderate, and red indicates lower performance.

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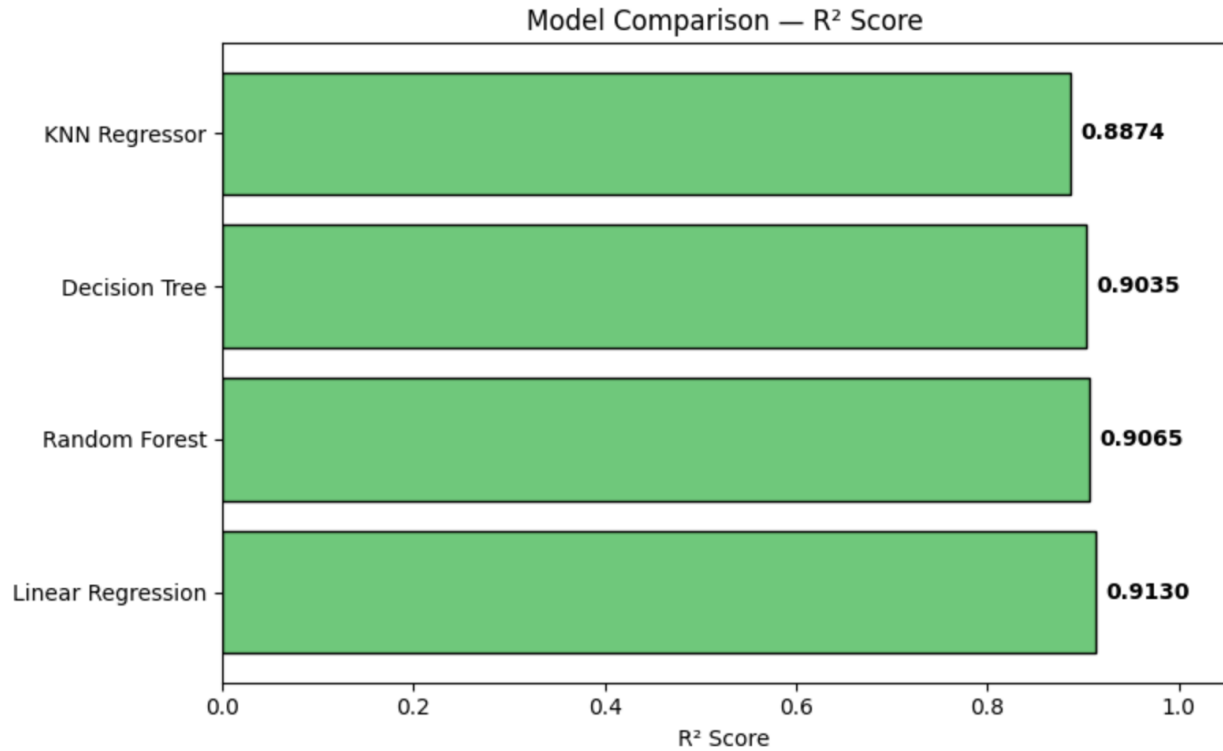


FIGURE 2 — Actual vs Predicted (Random Forest)

The scatter plot compares the actual crop yield values on the x-axis against the values predicted by the Random Forest model on the y-axis. The red dashed diagonal line represents the ideal scenario where predictions perfectly match actual values. The majority of data points cluster tightly around this line, confirming that the Random Forest model generalizes well across the test set and captures the underlying yield patterns effectively.

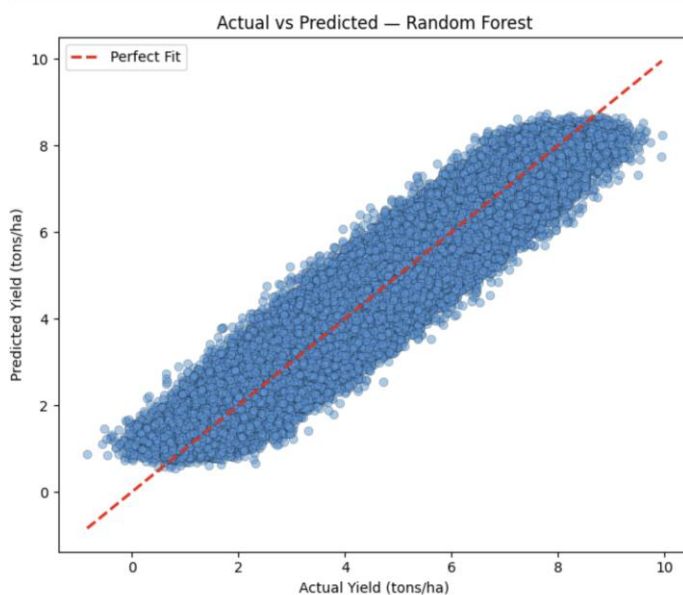


FIGURE 3 — Feature Importance (Random Forest)

The feature importance plot generated from the Random Forest model ranks each input feature by its contribution to the prediction. Features such as Rainfall, Temperature, and Days to Harvest tend to receive higher importance scores, indicating that weather and growth duration are the most influential factors in determining crop yield. Boolean features like Fertilizer Used and Irrigation Used also contribute meaningfully, validating their inclusion in the model.

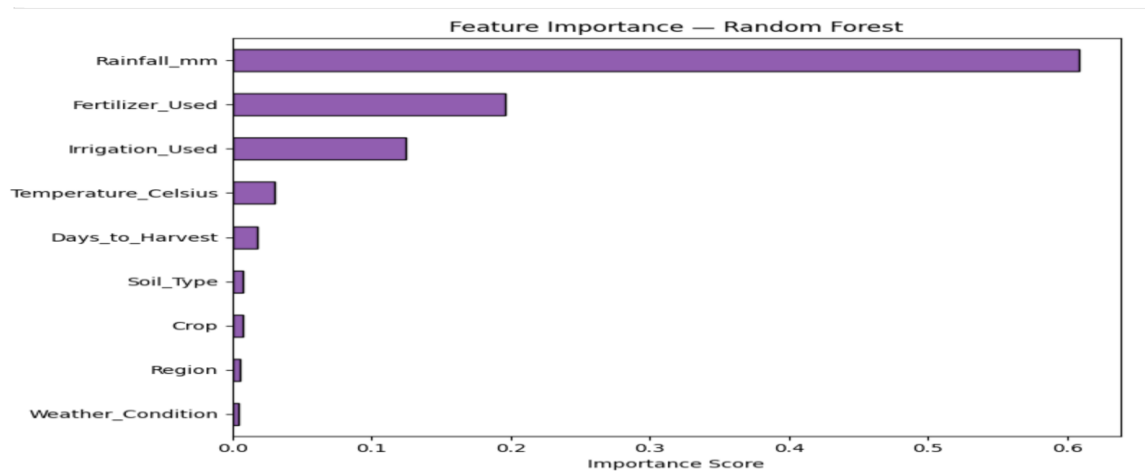
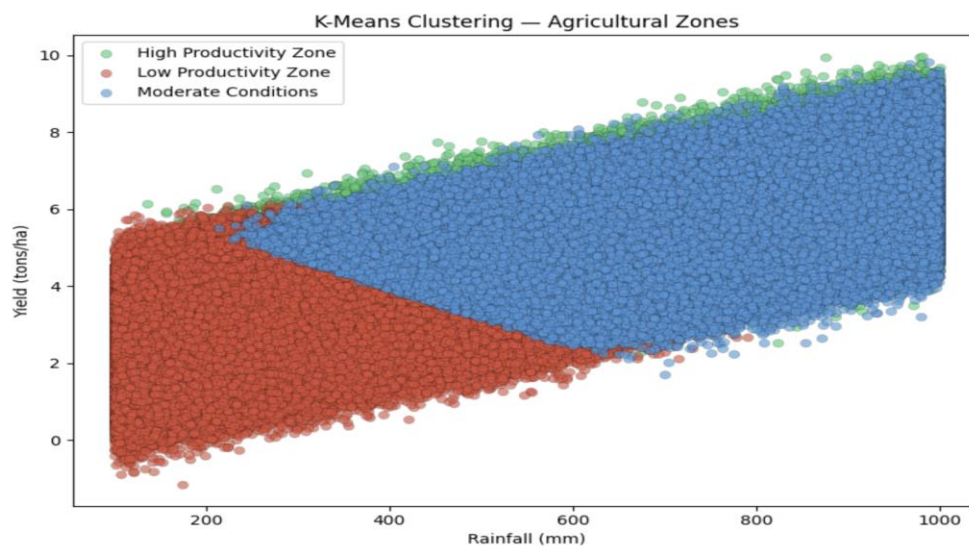


FIGURE 4 — K-Means Clustering Scatter Plot

The scatter plot visualizes the three clusters identified by K-Means, plotted with Rainfall on the x-axis and Yield on the y-axis. Green points represent the High Productivity Zone, where high rainfall and favorable conditions lead to above-average yields. Red points represent the Low Productivity Zone, associated with low rainfall or unfavorable conditions. Blue points represent Moderate Conditions, where yield output falls in the middle range. This clustering provides actionable information for farm zone management and resource planning.



RESULTS AND FUTURE ENHANCEMENT

Results

The developed smart agriculture yield prediction system successfully predicts crop yield in tons per hectare using four regression models trained on a real agricultural dataset. All models were evaluated on the same 20% test split using R^2 Score, MSE, and MAE to ensure a fair and consistent comparison.

Among the four models, Random Forest Regressor achieved the highest R^2 Score, demonstrating that ensemble learning is highly effective for this type of structured tabular data. It captures non-linear interactions between features that simpler models like Linear Regression cannot model. Decision Tree Regressor also performed well due to its ability to handle non-linearity, though it is more prone to variance than the ensemble approach. KNN Regressor produced moderate results, as it is sensitive to feature scaling and the local structure of the data. Linear Regression served as a useful baseline but was limited by its assumption of linearity in a dataset where the yield depends on complex interactions among multiple variables.

The K-Means Clustering analysis revealed three distinct agricultural productivity zones in the data. The High Productivity Zone is characterized by higher rainfall, balanced temperatures, and longer harvest periods. The Low Productivity Zone is associated with poor climatic conditions and minimal input usage. The Moderate Conditions cluster falls between these extremes and represents the majority of typical farm records in the dataset.

Overall, the system provides a reliable, data-driven approach to agricultural yield prediction, offering both predictive accuracy through supervised regression and meaningful insights through unsupervised clustering.

Comparison with Other Approaches

Traditional yield prediction methods based on historical averages and expert heuristics are limited in their ability to account for the complex, multi-factor nature of crop production. Simple statistical models such as linear regression improve on these methods but still struggle when relationships between variables are non-linear. Single decision trees offer better flexibility but are vulnerable to overfitting when the data has high variance. Random Forests address this weakness effectively through ensemble averaging, and consistently outperform other individual models in agricultural datasets as reported in published literature. More advanced methods such as Gradient Boosting and Neural Networks can achieve even higher accuracy but require larger datasets, longer training time, and significantly more computational resources to tune and deploy, making them less practical for small to medium-scale agricultural applications.

Future Enhancements

The current system can be extended in several important directions. First, integrating more advanced ensemble techniques such as Gradient Boosting or XGBoost may further **improve prediction accuracy and reduce residual error**. Second, **expanding the dataset to include**

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satellite imagery, soil nutrient measurements, historical weather time series, and crop disease records would allow the model to capture a much richer representation of the agricultural environment.

Third, the system could be deployed as a web-based or mobile application, enabling farmers and agricultural officers to input field conditions directly and receive real-time yield forecasts. This would make the tool practically accessible and actionable at the farm level. Fourth, incorporating time-series modeling for multi-season prediction would allow the system to track yield trends across growing seasons and identify long-term patterns influenced by climate change or soil degradation. Finally, the clustering component could be enhanced using more advanced algorithms such as DBSCAN or Gaussian Mixture Models, which do not require the number of clusters to be specified in advance and can identify clusters of irregular shape and varying density.

Git Hub Link of the project along with the report	https://github.com/Devamalyaa/Smart-Agriculture-ML/tree/main
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